

# THE CONFABULATION ENGINE

*Why Artificial Intelligence Does Not Lie — It Confabulates*

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*August 2026*

**Abstract**—When large language models generate false assertions with pristine confidence, the prevailing public and technical moniker is 'hallucination.' Borrowed loosely from clinical psychiatry, the term is fundamentally misapplied. Hallucination presupposes a perceiving subject whose sensory channels misfire; artificial neural networks possess no sensory organs, no perceptual continuum, and no subjective awareness to experience a phantom percept. This paper proposes a far more precise and historically grounded conceptual framework: confabulation. Originating in late nineteenth-century neuropsychiatry and refined through twentieth-century split-brain research, confabulation designates the fluent, unhesitating generation of a false narrative by a cognitive system structurally incapable of recognizing its own ignorance. Drawing upon Michael Gazzaniga and Joseph LeDoux's discovery of the cerebral 'interpreter,' alongside the clinical lineage extending from Karl Bonhoeffer to William Hirstein, we demonstrate that modern generative language models operate as architectural confabulation engines. They are built to generate continuous, syntactically coherent completions without any internal mechanism to cross-examine output against objective reality. We evaluate this framework against alternative accounts—including Emily Bender et al.'s 'stochastic parrot' thesis and Harry Frankfurt's concept of 'bullshit' (as recently applied by Hicks et al.)—and examine empirical evidence from legal filings and comparative output analysis. We conclude by outlining the institutional and structural safeguards necessary when deploying technologies whose primary vulnerability lies in their capacity to mimic the cadence of authentic knowledge.

**Keywords:** *confabulation, artificial intelligence, large language models, epistemology, philosophy of mind, Gazzaniga, interpreter theory*

# 1. Introduction

Since the widespread public deployment of large-scale generative language systems, the term 'hallucination' has served as the default label for their factual errors. Yet the metaphor is misleading in ways that misdirect both technical remediation and public understanding. In clinical neurology and psychiatry, a hallucination is a sensory perception occurring in the absence of an external stimulus—a visual apparition, an auditory command, or a phantom touch experienced by a conscious mind. Large language models possess no perceptual apparatus, no sensory inputs to distort, and no inner phenomenal space in which a perception could materialize. To speak of a model 'hallucinating' inadvertently imports a suite of anthropomorphic assumptions regarding awareness, perception, and attentional focus that bears no relation to modern transformer architectures.

This paper argues for a different, richer, and historically grounded term: confabulation. In clinical neuropsychiatry, confabulation describes the production of false statements or fabricated memories without any conscious intention to deceive, occurring in individuals who genuinely believe in the truth of their assertions. The confabulating patient is neither lying (which requires knowing the truth and choosing to conceal it) nor hallucinating (which requires phantom sensory input). Instead, confabulation emerges when a system tasked with producing coherent narrative completes a gap in knowledge with plausible, contextually continuous detail—unaware that the fabric of its response is entirely invented.

By examining the architectural properties of generative language models through the lens of cognitive neurology, we can see that these systems are not reliable factual databases that occasionally suffer a sensory glitch. Rather, they are architectural confabulation engines: systems engineered to maintain seamless textual fluency regardless of whether the underlying facts are present, verified, or entirely absent.

## 2. Confabulation: A Concept from the Clinic

The clinical genealogy of confabulation provides crucial insights into how complex information systems respond to internal gaps. The concept first took formal shape in Karl Bonhoeffer's 1901 investigations into alcoholic psychoses [1] and Sergei Korsakoff's earlier clinical descriptions of severe amnesic syndromes. As the medical historian German Berrios has documented, early twentieth-century alienists recognized that confabulating patients were not engaged in deliberate prevarication [2]. When asked where they had been the previous evening, patients with Korsakoff's syndrome would describe a detailed dinner party or a walk in the park with complete conviction, utterly unaware that they had been confined to a hospital ward for months. What distinguished these accounts from ordinary lies was the total absence of felt hesitation or deceptive intent; the narrative arrived with the felt authority of authentic memory.

The neurological mechanics of this phenomenon were dramatically illuminated in the late 1970s by Michael Gazzaniga and Joseph LeDoux during their pioneering studies of split-brain patients [3]. In these individuals, the corpus callosum had been surgically severed to control intractable epilepsy, functionally isolating the left and right cerebral hemispheres. In one classic experiment, separate visual stimuli were presented simultaneously to each hemisphere: the right hemisphere (controlling the left hand) saw a winter snow scene, while the left hemisphere (controlling speech and the right hand) saw a chicken claw [3]. When asked to choose matching images from a set array, the patient's left hand selected a snow shovel (matching the snow scene) and his right hand selected a chicken (matching the claw). When the experimenter asked the patient why his left hand was pointing to a shovel, his left hemisphere—which

governed speech but had no access to the snow scene shown to the right hemisphere—did not admit ignorance. Instead, without a moment's pause, the patient's verbal left hemisphere declared: 'Oh, that's simple. The chicken claw goes with the chicken, and you need a shovel to clean out the chicken shed' [5].

Gazzaniga termed the neural module responsible for this spontaneous narrative synthesis 'the interpreter' [4]. The interpreter's mandate is to construct a plausible, causally coherent explanation for observed behaviors and internal states, even when it lacks access to the actual underlying data. Subsequent cognitive research revealed that the interpreter is not merely an artifact of brain surgery, but a ubiquitous, baseline feature of human cognition—the very system that allows individuals to construct post-hoc rationalizations for unconscious impulses and reflexive actions [5].

Philosophical and neuropsychiatric analysis further clarifies where confabulation sits relative to other speech modes. The liar possesses the truth and intentionally obscures it. Harry Frankfurt, in his seminal work *On Bullshit*, identified a third category: the bullshitter, who is indifferent to truth-value altogether, manipulating language purely to persuade or impress [6]. However, as William Hirstein has argued in *Brain Fiction*, confabulation represents a distinct structural failure [7]. Unlike the bullshitter, who possesses the capacity to track truth but deliberately chooses to disregard it, the confabulating agent lacks the internal monitoring apparatus required to verify its claims against reality. Confabulation is a breakdown in the system's self-checking mechanism, resulting in the confident delivery of ungrounded narrative.

### **3. Why the Machine Confabulates**

To understand why generative artificial intelligence exhibits this exact behavior, one must look directly at transformer architecture. A large language model is trained to predict the statistically probable continuation of a sequence of tokens across a vast corpus of human language. In their celebrated thought experiment, Emily Bender and Alexander Koller illustrated the fundamental constraint of learning strictly from surface form [8]. They imagined a hyper-intelligent octopus intercepting telegraph messages transmitted across an underwater cable between two isolated humans. By analyzing statistical patterns across millions of messages, the octopus could learn to construct responses that perfectly mimic human dialogue, yet it would remain entirely devoid of real-world reference, sensory grounding, or semantic intent [8]. Building on this critique, Bender, Gebru, McMillan-Major, and Shmitchell famously described language models as 'stochastic parrots'—systems capable of synthesizing complex linguistic form without any underlying comprehension of fact or reference [9].

This structural disconnection between linguistic form and empirical reality creates the exact conditions for programmatic confabulation. As clinicians Andrew Smith, Felix Greaves, and Trishan Panch have argued, applying the term 'hallucination' to language models is a metaphorical error that obscures their true operation [10]. A model possesses no perceptual awareness to hallucinate; rather, when queried on a topic where training data is sparse or absent, it synthesizes plausible syntax and domain vocabulary to fulfill its statistical objective. It generates narrative detail that is factually false without possessing any internal signal that an error has occurred.

The structural alignment between Gazzaniga's cerebral interpreter and a large language model is striking. Both systems are architecturally programmed to respond continuously—never to remain silent. A base language model, prior to fine-tuning, evaluates output solely on statistical likelihood; in human written text, a direct, confident answer is exponentially more probable than an explicit refusal or a nuanced

confession of ignorance. Furthermore, neither the split-brain interpreter nor the autoregressive language model contains an internal validation module that verifies a proposition against an external reality engine prior to output delivery. In split-brain patients, the connection to the opposite hemisphere is physically severed; in language models, no part of the neural network architecture is dedicated to truth-verification. Confabulation is not an episodic bug; it is the default operational state of an unconstrained narrative generator.

#### **4. Empirical Evidence and Alternative Frameworks**

The real-world consequences of architectural confabulation are no longer theoretical. In 2023, during federal litigation in the United States (*Mata v. Avianca*), an attorney utilized a commercial language model to draft a legal brief without cross-referencing its legal citations [11]. The resulting document cited multiple judicial precedents complete with elaborate case summaries, procedural histories, and direct judicial quotations—none of which existed in any legal database [11]. The presiding judge observed that the filing contained fictitious judicial decisions supported by entirely fabricated quotations. Crucially, these false citations were not random gibberish or chaotic noise; they were impeccably formatted, adopted authentic legal nomenclature, and exhibited the exact structural features of genuine judicial opinions.

This phenomenon reflects a broader, measurable pattern across generative models. In a 2024 empirical study, Peiqi Sui and colleagues conducted a systematic analysis of language model outputs that were independently verified as false [12]. Their findings revealed a startling dynamic: false model outputs were, on average, measurably more narratively coherent, semantically rich, and syntactically persuasive than the models' true statements [12]. This directly mirrors clinical findings in neuropsychiatry, where confabulated memories often possess greater vividness and narrative completeness than genuine, fragmented recollections. When freed from the constraints of strict factual correspondence, the generative network's narrative capacity operates at maximum output density.

This structural reality allows us to evaluate competing philosophical interpretations of model errors. Michael Townsen Hicks, James Humphries, and Joe Slater have argued that large language models are best understood through Harry Frankfurt's concept of 'bullshit,' asserting that the systems are fundamentally indifferent to truth [13]. In their analysis, Hicks et al. explicitly reject the term confabulation—attributing its popularization to tech journalist Benj Edwards [14]—on the grounds that it risks anthropomorphizing machine code and implies that errors are exceptional events rather than products of the same underlying generative process that produces accurate text [13, 14].

However, a deeper reading of Gazzaniga's clinical literature actually resolves Hicks et al.'s objection and reinforces the confabulation framework. Gazzaniga emphasized that the cerebral interpreter does not switch on exclusively when a patient makes a mistake; the interpreter operates continuously, generating post-hoc narratives for both true and false behaviors through the exact same neurological process [4, 5]. Confabulation is simply the term applied when the interpreter's continuous narrative generation fails to correspond with external reality. The same principle applies to language models: a single, uniform statistical mechanism generates both accurate outputs and fabricated ones. The term confabulation correctly identifies what occurs when that generative process operates without a truth-tracking constraint, whereas Frankfurian 'indifference' presupposes an intentional agent capable of tracking truth who deliberately chooses to ignore it—a capacity generative networks simply do not possess.

## 5. What Follows: Implications and Institutional Safeguards

Reframing machine error as confabulation shifts the discourse away from speculative questions about machine consciousness or deceptive intent toward actionable system design and institutional accountability. If confabulation is an inherent property of autoregressive language generation, several critical implications follow directly:

- 1. Technical and Regulatory Terminology:** Technical standards, regulatory frameworks, and public policy documents should abandon the misleading terminology of 'hallucination' in favor of 'confabulation.' Recognizing that errors stem from unconstrained narrative generation rather than sensory distortion clarifies the precise nature of the technical challenge.
- 2. Calibrated Uncertainty as a Core Requirement:** Generative architectures must be evaluated on their ability to signal uncertainty and acknowledge knowledge boundaries. Developing models that reliably output calibrated confidence intervals or express ignorance when data is missing is a foundational prerequisite for safe deployment.
- 3. Architectural Verification Outer-Loops:** Because the generative core lacks internal truth-tracking, external verification mechanisms must be built around it. Retrieval-Augmented Generation (RAG), formal logic checkers, and deterministic tool integration act as essential prosthetics—analogue to memory aids and external logs provided to clinical patients—supplying the empirical grounding that the autoregressive model cannot provide for itself.
- 4. High-Stakes Verification Protocols:** In domains where factual accuracy is paramount—including medicine, law, engineering, and journalism—institutional policies must mandate independent human verification by design. Relying on model fluency as a proxy for factual correctness represents a fundamental failure of institutional governance.

## 6. Conclusion

The factual errors of generative artificial intelligence are neither mystery nor sensory illusion. They represent the predictable output of a confabulation engine: a system engineered for seamless fluency, continuous generation, and contextual plausibility, yet devoid of any internal capacity to cross-examine its claims against reality. The evidence from clinical neurology, philosophical analysis, legal proceedings, and empirical text evaluation converges on a single conclusion: fluency is not truth. Until external architectures of verification and rigorous human oversight are embedded around these systems, deploying them in high-stakes environments remains a dangerous surrender to manufactured certainty.

## References

1. Bonhoeffer, K. (1901). Die akuten Geisteskrankheiten der Gewohnheitstrinker [The acute mental disorders of habitual drinkers]. Fischer.
2. Berrios, G. E. (1998). Confabulations: A conceptual history. *Journal of the History of the Neurosciences*, 7(3), 225–241. <https://doi.org/10.1076/jhin.7.3.225.1855>
3. Gazzaniga, M. S., & LeDoux, J. E. (1978). *The integrated mind*. Plenum Press.
4. Gazzaniga, M. S. (2000). Cerebral specialization and interhemispheric communication: Does the corpus callosum enable the human condition? *Brain*, 123(7), 1293–1326. <https://doi.org/10.1093/brain/123.7.1293>
5. Gazzaniga, M. S. (2011). *Who's in charge? Free will and the science of the brain*. Ecco.
6. Frankfurt, H. G. (2005). *On bullshit*. Princeton University Press.
7. Hirstein, W. (2005). *Brain fiction: Self-deception and the riddle of confabulation*. MIT Press.
8. Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On meaning, form, and understanding in the age of data. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 5185–5198). <https://doi.org/10.18653/v1/2020.acl-main.463>
9. Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)* (pp. 610–623). <https://doi.org/10.1145/3442188.3445922>
10. Smith, A. L., Greaves, F., & Panch, T. (2023). Hallucination or confabulation? Neuroanatomy as metaphor in large language models. *PLOS Digital Health*, 2(11), e0000388. <https://doi.org/10.1371/journal.pdig.0000388>
11. Weiser, B. (2023, May 27). Here's what happens when your lawyer uses ChatGPT. *The New York Times*. <https://www.nytimes.com/2023/05/27/nyregion/avianca-airline-lawsuit-chatgpt.html>
12. Sui, P., Duede, E., Wu, S., & So, R. J. (2024). Confabulation: The surprising value of large language model hallucinations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Long Papers)* (pp. 14274–14284). <https://arxiv.org/abs/2406.04175>
13. Hicks, M. T., Humphries, J., & Slater, J. (2024). ChatGPT is bullshit. *Ethics and Information Technology*, 26(2), Article 38. <https://doi.org/10.1007/s10676-024-09775-5>
14. Edwards, B. (2023, April 6). Why ChatGPT and Bing Chat are so good at making things up. *Ars Technica*. <https://arstechnica.com/information-technology/2023/04/why-ai-chatbots-are-the-ultimate-bs-machines-and-how-people-hope-to-fix-them/>